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NANOPHOTONIC HYPERSPECTRAL IMAGING

Abstract

Image sensor pixels and methods for constructing the same. The image sensor pixel includes a first photodetector, a second photodetector, a first spectral router, and a second spectral router. The first spectral router is positioned above the first photodetector and the second photodetector. The first spectral router is configured to direct incident light in a first subset of a first color wavelength range to the first photodetector and direct incident light in a second subset of the first color wavelength range to the second photodetector. The second spectral router is positioned above the first spectral router. The second spectral router is configured to direct the incident light in the first subset and the second subset of the first color wavelength range to the first spectral router. The second spectral router is configured to direct incident light in a second color wavelength range to one or more neighboring image sensor pixels.

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Background/Summary

BACKGROUND

[0001] Image sensors are used in electronic devices such as cellular telephones, cameras, and computers to capture images. In particular, an electronic device is provided with an array of image sensor pixels arranged in a grid pattern. Each image sensor pixel receives incident photons, such as light, and converts the photons into electrical signals. Column circuitry is coupled to each column for reading out sensor signals from each image sensor pixel.

SUMMARY

[0002] For each pixel in an image, a hyperspectral camera may acquire the light intensity for a large number of contiguous spectral bands (typically a few tens to several hundred). Every pixel in the image thus contains a continuous spectrum and can be used to characterize the objects in the scene with great precision and detail. The present disclosure provides image sensor pixels, image sensors, and methods that, among other things, provide hyperspectral imaging using nanophotonic light guides (or spectral routers) to split color wavelength bands into sub-bands.

[0003] The present disclosure provides an image sensor pixel including, in one implementation, a plurality of photodetectors, a first spectral router, and a second spectral router. The plurality of photodetectors includes at least a first photodetector and a second photodetector. The first spectral router is positioned above at least the first photodetector and the second photodetector. The first spectral router is configured to direct incident light in a first subset of a first color wavelength range to the first photodetector. The first spectral router is also configured to direct incident light in a second subset of the first color wavelength range to the second photodetector. The second spectral router is positioned above the first spectral router. The second spectral router is configured to direct the incident light in the first subset and the second subset of the first color wavelength range to the first spectral router. The second spectral router is configured to direct incident light in a second color wavelength range to one or more neighboring image sensor pixels.

[0004] The present disclosure also provides an image sensor pixel including, in one implementation, a pixel array and a controller. The pixel array includes an image sensor pixel. The image sensor pixel includes a first photodetector, a second photodetector, a first spectral router, and a second spectral router. The first spectral router is configured to direct incident light in a first subset of a first color wavelength range to the first photodetector. The first spectral router is also configured to direct incident light in a second subset of the first color wavelength range to the second photodetector. The second spectral router is configured to direct incident light in the first color wavelength range to the first spectral router. The second spectral router is also configured to direct incident light in a second color wavelength range to one or more neighboring image sensor pixels in the pixel array. The controller is configured to determine a first amount of light detected by the first photodetector. The controller is also configured to determine a second amount of light detected by the second photodetector.

[0005] The present disclosure further provides a method for constructing an image sensor pixel. The method includes forming a plurality of photodetectors including at least a first photodetector and a second photodetector. The method also includes forming a first spectral router positioned above at least the first photodetector and the second photodetector. The first spectral router is configured to direct incident light in a first subset of a first color wavelength range to the first photodetector. The first spectral router is also configured to direct incident light in a second subset of the first color wavelength range to the second photodetector. The method further includes forming a second spectral router positioned above the first spectral router. The second spectral router is configured to direct the incident light in the first subset and the second subset of the first color wavelength range to the first spectral router. The second spectral router is also configured to

direct incident light in a second color wavelength range to one or more neighboring image sensor pixels.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] For a detailed description of example implementations, reference will now be made to the accompanying drawings in which:

[0007] FIG. 1A is a block diagram of an example of an imaging system in accordance with some implementations;

[0008] FIG. 1B is a diagram of an example of an imaging system incorporated in a vehicle in accordance with some implementations;

[0009] FIG. 2 is a partial schematic and a partial block diagram of an example of an image sensor in accordance with some implementations;

[0010] FIG. 3 is a schematic of an example of circuitry in an image sensor pixel in accordance with some implementations;

[0011] FIG. 4A is a view of an example of a color pattern of split-pixels in accordance with some implementations;

[0012] FIG. 4B is a cross-sectional view of the color pattern of FIG. 4A in accordance with some implementations in accordance with some implementations;

[0013] FIG. 5A is a graph of an example of the red light wavelength range divided into four wavelength subsets in accordance with some implementations;

[0014] FIG. 5B is a view of an example of a red split-pixel configured to detect four wavelength subsets in accordance with some implementations;

[0015] FIG. 6A is a graph of an example of the red light wavelength range divided into two wavelength subsets in accordance with some implementations;

[0016] FIG. 6B is a view of an example of a diagonally arranged red split-pixel configured to detect two wavelength subsets in accordance with some implementations;

[0017] FIG. 6C is a view of an example of a horizontally arranged red split-pixel configured to detect two wavelength subsets in accordance with some implementations;

[0018] FIG. 6D is a view of an example of a vertically arranged red split-pixel configured to detect two wavelength subsets in accordance with some implementations; and

[0019] FIG. 7 is a flow diagram of an example of a method for constructing an image sensor pixel in accordance with some implementations.

DEFINITIONS

[0020] Various terms are used to refer to particular system components. Different companies may refer to a component by different names—this document does not intend to distinguish between components that differ in name but not function. In the following discussion and in the claims, the terms “including” and “comprising” are used in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to . . .” Also, the term “couple” or “couples” is intended to mean either an indirect or direct connection. Thus, if a first device couples to a second device, that connection may be through a direct connection or through an indirect connection via other devices and connections.

[0021] Terms defining an elevation, such as “above,” “below,” “upper”, and “lower” shall be locational terms in reference to a direction of light incident upon a pixel array and/or an image pixel. Light entering shall be considered to interact with or pass objects and/or structures that are “above” and “upper” before interacting with or passing objects and/or structures that are “below” or “lower.” Thus, the locational terms may not have any relationship to the direction of the force of gravity.

[0022] “A”, “an”, and “the” as used herein refers to both singular and plural referents unless the context clearly dictates otherwise. By way of example, “a processor” programmed to perform various functions refers to one processor programmed to perform each and every function, or more than one processor collectively programmed to perform each of the various functions. To be clear, an initial reference to “a [referent]”, and then a later reference for antecedent basis purposes to “the [referent]”, shall not obviate the fact the recited referent may be plural.

[0023] In relation to electrical devices, whether stand alone or as part of an integrated circuit, the terms “input” and “output” refer to electrical connections to the electrical devices, and shall not be read as verbs requiring action. For example, a differential amplifier, such as an operational amplifier, may have a first differential input and a second differential input, and these “inputs” define electrical connections to the operational amplifier, and shall not be read to require inputting signals to the operational amplifier.

[0024] “Light” or “color” shall mean visible light ranging between about 380 and 700 nanometers. “Light” or “color” shall also mean light ranging between 700 nanometers to 800 nanometers, and invisible light, such as infrared light ranging between about 800 nanometer and 1 millimeter. “Light” or “color” shall also mean invisible light, such as ultraviolet light ranging between about 100 nanometers to 400 nanometers.

[0025] “Controller” shall mean, alone or in combination, individual circuit components, an application specific integrated circuit (ASIC), one or more microcontrollers with controlling software, a reduced-instruction-set computer (RISC) with controlling software, a digital signal processor (DSP), one or more processors with controlling software, a programmable logic device (PLD), a field programmable gate array (FPGA), or a programmable system-on-a-chip (PSOC), configured to read inputs and drive outputs responsive to the inputs.

DETAILED DESCRIPTION

[0026] The following discussion is directed to various implementations of the invention. Although one or more of these implementations may be preferred, the implementations disclosed should not be interpreted, or otherwise used, as limiting the scope of the present disclosure, including the claims. In addition, one skilled in the art will understand that the following description has broad application, and the discussion of any implementation is meant only to be exemplary of that implementation, and not intended to intimate that the scope of the present disclosure, including the claims, is limited to that implementation.

[0027] Various examples are directed to image sensor pixels, image sensors, and related methods. More particularly, at least some examples are directed to image sensor pixels and imaging systems that provide hyperspectral imaging. More particular still, various examples are directed to split-pixels that use nanophotonic light guides (or spectral routers) to split color wavelength bands into sub-bands. The specification now turns to an example system to orient the reader.

[0028] FIG. 1A shows an example of an imaging system **100**. In particular, the imaging system **100** may be a portable electronic device such as a camera, a cellular telephone, a tablet computer, a webcam, a video camera, a video surveillance system, or a video gaming system with imaging capabilities. In other cases, the imaging system **100** may be an automotive imaging system. The imaging system **100** illustrated in FIG. 1A includes a camera module **102** that may be used to convert incoming light into digital image data. The camera module **102** may include one or more lenses **104** and one or more corresponding image sensors **106**. The lenses **104** may include fixed and/or adjustable lenses. During image capture operations, light from a scene may be focused onto the image sensor **106** by the lenses **104**. The image sensor **106** may comprise circuitry for converting analog pixel data into corresponding digital image data to be provided to the imaging controller **108**. If desired, the camera module **102** may be provided with an array of lenses **104** and an array of corresponding image sensors **106**.

[0029] The imaging controller **108** may include one or more integrated circuits. The imaging circuits may include image processing circuits, microprocessors, and storage devices, such as

random-access memory, and non-volatile memory. The imaging controller **108** may be implemented using components that are separate from the camera module **102** and/or that form part of the camera module **102**, for example, circuits that form part of the image sensor **106**. Digital image data captured by the camera module **102** may be processed and stored using the imaging controller **108**. Processed image data may, if desired, be provided to external equipment, such as computer, external display, or other device, using wired and/or wireless communications paths coupled to the imaging controller **108**.

[0030] FIG. **1B** shows another example of the imaging system **100**. The imaging system **100** illustrated in FIG. **1B** comprises an automobile or vehicle **110**. The vehicle **110** is illustratively shown as a passenger vehicle, but the imaging system **100** may be other types of vehicles, including commercial vehicles, on-road vehicles, and off-road vehicles. Commercial vehicles may include busses and tractor-trailer vehicles. Off-road vehicles may include tractors and crop harvesting equipment. In the example of FIG. **1B**, the vehicle **110** includes a forward-looking camera module **102** arranged to capture images of scenes in front of the vehicle **110**. Such forward-looking camera module **102** can be used for any suitable purpose, such as lane-keeping assist, collision warning systems, distance-pacing cruise-control systems, autonomous driving systems, and proximity detection. The vehicle **110** further comprises a backward-looking camera module **102** arranged to capture images of scenes behind the vehicle **110**. Such backward-looking camera module **102** can be used for any suitable purpose, such as collision warning systems, reverse direction video, autonomous driving systems, proximity detection, monitoring position of overtaking vehicles, and backing up. The vehicle **110** may further comprise an inside-looking camera module **102** arranged to capture images of scenes inside the vehicle **110**. Such inside-looking camera module **102** can be used for any suitable purpose, such as an in-cabin Driver Monitoring System (DMS) and a Driver plus Occupant Monitoring System (DOMS). The vehicle **110** further comprises a side-looking camera module **102** arranged to capture images of scenes beside the vehicle **110**. Such side-looking camera module can be used for any suitable purpose, such as blind-spot monitoring, collision warning systems, autonomous driving systems, monitoring position of overtaking vehicles, lane-change detection, and proximity detection. In situation in which the imaging system **100** is a vehicle, the imaging controller **108** may be a controller of the vehicle **110**. The discussion now turns in greater detail to the image sensor **106** of the camera module **102**.

[0031] FIG. **2** shows an example of the image sensor **106**. In particular, FIG. **2** shows that the image sensor **106** may comprise a substrate **200** of semiconductor material (for example, silicon) encapsulated within packaging to create a packaged semiconductor device or packaged semiconductor product. Bond pads or other connection points of the substrate **200** couple to terminals of the image sensor **106**, such as the serial communication channel **202** coupled to terminal(s) **204**, and capture input **206** coupled to terminal **208**. Additional terminals will be present, such as ground, common, or power, but the additional terminals are omitted so as not to unduly complicate the figure. While a single instance of the substrate **200** is shown, in other cases multiple substrates may be combined to form the image sensor **106** to form a multi-chip module.

[0032] The image sensor **106** comprises a pixel array **210** containing a plurality of image sensor pixels **212** arranged in rows and columns. Pixel array **210**, being one example of an “array of pixels,” may comprise, for example, hundreds or thousands of rows and columns of image sensor pixels **212**. Control and readout of the pixel array **210** may be implemented by an image sensor controller **214** coupled to a row controller **216** and a column controller **218**. The row controller **216** may receive row addresses from image sensor controller **214** and supply corresponding row control signals to image sensor pixels **212**, such as reset, row-select, charge transfer, dual conversion gain, and readout control signals. The row control signals may be communicated over one or more conductors, such as row control paths **220**.

[0033] Column controller **218** may be coupled to the pixel array **210** by way of one or more

conductors, such as column lines **222**. Column controllers may sometimes be referred to as column control circuits, readout circuit, or column decoders. Column lines **222** may be used for reading out image signals from image sensor pixels **212** and for supplying bias currents and/or bias voltages to image sensor pixels **212**. If desired, during pixel readout operations, a pixel row in the pixel array **210** may be selected using row controller **216** and image signals generated by image sensor pixels **212** in that pixel row can be read out along column lines **222**.

[0034] The column controller **218** may include sample-and-hold circuitry for sampling and temporarily storing image signals read out from pixel array **210**, amplifier circuitry, analog-to-digital conversion (ADC) circuitry, bias circuitry, column memory, latch circuitry for selectively enabling or disabling the column circuitry, or other circuitry that is coupled to one or more columns of pixels in the pixel array **210** for operating the image sensor pixels **212** and for reading out image signals from the image sensor pixels **212**. ADC circuitry in the column controller **218** may convert analog pixel values received from the pixel array **210** into corresponding digital image data.

Column controller **218** may supply digital image data to the image sensor controller **214** and/or the imaging controller **108** (FIG. **1A**) over, for example, the serial communication channel **202**.

[0035] FIG. **3** shows an example of circuitry in one of the image sensor pixels **212**. The image sensor pixels **212** may have fewer components, additional components, or different components in different configurations than the one illustrated in FIG. **3**. In particular, FIG. **3** shows that each of the image sensor pixels **212** may comprise a photodetector **302** (for example, a photodiode). A positive pixel power supply voltage, such as supply voltage V_{APIX}, may be supplied at a positive power supply terminal **304**. A ground power supply voltage, such a reference voltage V_{ss}, may be supplied at a ground terminal **306**. Incoming light is gathered by the photodetector **302**. The photodetector **302** converts the light to electrical charge.

[0036] Before an image is acquired, a reset control signal RST may be asserted. The reset control signal RST turns on a reset transistor **308** and resets a charge storage (CS) node **310** to a voltage equal or close to the supply voltage V_{APIX}. The reset control signal RST may then be de-asserted to turn off the reset transistor **308**. After the reset process is complete, a transfer gate control signal TX may be asserted to turn on a transfer transistor **312**. When the transfer transistor **312** is turned on, charge generated by the photodetector **302** in response to incoming light is transferred to the charge storage node **310**. The charge storage node **310** exhibits a capacitance that can be used to store the charge that has been transferred from the photodetector **302**. The signal associated with the charge stored in the charge storage node **310** is buffered by a source-follower transistor **314**. A row select transistor **316** connects the source-follower transistor **314** to one of the column lines **222**.

[0037] When it is desired to read out the value of the charge stored in the charge storage node **310**, a control signal RS is asserted. The read-out value may be, for example, the value of the charge storage node **310** that is represented by the signal at the source terminal S of the source-follower transistor **314**. When the control signal RS is asserted, the row select transistor **316** is turned on and an output signal V_{out} that is representative of the magnitude of the charge stored in the charge storage node **310** is produced on an output path **318**. The output signal V_{out} is one example of a “pixel signal.” When the control signal RS is asserted, one of the column lines **222** can be used to route the output signal V_{out} from the image sensor pixel **212** to readout circuitry, such as the column controller **218** in FIG. **2**.

[0038] FIG. **4A** shows a view of an example of a color pattern **400** of split-pixels. In particular, the color pattern **400** includes a red split-pixel **402**, a first green split-pixel **404**, a second green split-pixel **406**, and a blue split-pixel **408** arranged in a two-by-two grid. Other grid patterns are also possible, such as one-by-four grid. Other color patterns are possible, such as red-yellow-yellow-cyan. The red split-pixel **402** is one example of an “image sensor pixel.” The red split-pixel **402** shown in FIG. **4A** includes a first red photodetector **410**, a second red photodetector **412**, a third red photodetector **414**, and a fourth red photodetector **416** arranged in a two-by-two grid. In the

example shown, the four photodetectors of the red split-pixel **402** abut each other, but in other cases one or more additional layers, such as oxide layers or deep trench isolation (DTI) structures, may reside between them. The first green split-pixel **404** shown in FIG. **4A** includes a first green photodetector **418**, a second green photodetector **420**, a third green photodetector **422**, and a fourth green photodetector **424** arranged in a two-by-two grid. A similar discussion regarding the second green split-pixel **406** and the blue split-pixel **408**, each of which may be configured in a same or similar manner, is omitted so as not to unduly lengthen the specification.

[0039] FIG. **4B** shows a cross-sectional view of the color pattern **400** taken at line **4B-4B** of FIG. **4A**. In particular, FIG. **4B** shows that the red split-pixel **402** further includes a red spectral router **426** positioned above the first red photodetector **410** and the second red photodetector **412**. The first red photodetector **410** is one example of a “first photodetector.” The second red photodetector **412** is one example of a “second photodetector.” Although not visible in the cross-sectional view of FIG. **4B**, the red spectral router **426** is also positioned above the third red photodetector **414** and the fourth red photodetector **416**. The third red photodetector **414** is one example of a “third photodetector.” The fourth red photodetector **416** is one example of a “fourth photodetector.” In the example shown in FIG. **4B**, the red spectral router **426** abuts the first red photodetector **410** and the second red photodetector **412**, but in other cases one or more additional layers, such as oxide or planar layers, may reside between them.

[0040] FIG. **4B** also shows that the first green split-pixel **404** further includes a green spectral router **428** positioned above the first green photodetector **418** and the second green photodetector **420**. The first green photodetector **418** is one example of a “third photodetector.” The second green photodetector **420** is one example of a “fourth photodetector.” Although not visible in the cross-sectional view of FIG. **4B**, the green spectral router **428** is also positioned above the third green photodetector **422** and the fourth green photodetector **424**. In the example shown in FIG. **4B**, the green spectral router **428** abuts the first green photodetector **418** and the second green photodetector **420**, but in other cases one or more additional layers, such as oxide or planar layers, may reside between them.

[0041] FIG. **4B** also shows a spectral router **430** is positioned above the red spectral router **426** and the green spectral router **428**. In the example shown in FIG. **4B**, the spectral router **430** abuts the red spectral router **426** and the green spectral router **428**, but in other cases one or more additional layers, such as oxide or planar layers, may reside between them.

[0042] A spectral router (or nanophotonic light guide) is an optical structure that accepts photons incident on an upper surface. The spectral router then diverts photons from the upper surface to the underlying photosensitive regions of photodiodes. The spectral router **430** is one example of a “second spectral router.” The spectral router **430** is configured to direct incident light in the red wavelength range to the red spectral router **426**. For example, the portion of the spectral router **430** positioned above the red split-pixel **402** is configured to pass incident light in the red wavelength range, such as between about 625 nanometers and 750 nanometers. Consider, for purposes of discussion, red light entering the red split-pixel **402**. An example of red light is illustrated in FIG. **4B** by arrow **432**. The red light initially encounters a portion of the spectral router **430** positioned above the red split-pixel **402**, which passes the red light to the red spectral router **426** of the red split-pixel **402**. Further, a portion of the spectral router **430** positioned above the first green split-pixel **404** is configured to direct incident light in the red wavelength range to the red spectral router **426** of the red split-pixel **402**. Consider, for purposes of discussion, red light entering the first green split-pixel **404**. An example of red light is illustrated in FIG. **4B** by arrow **434**. The red light initially encounters a portion of the spectral router **430** positioned above of the first green split-pixel **404**, which directs the red light to the red spectral router **426** of the red split-pixel **402**.

[0043] The spectral router **430** is also configured to direct incident light in the green wavelength range to the green spectral router **428**. For example, the portion of the spectral router **430** positioned above the first green split-pixel **404** is configured to pass incident light in the green

wavelength range, such as between about 495 nanometers and 570 nanometers. Consider, for purposes of discussion, green light entering the first green split-pixel **404**. An example of green light is illustrated in FIG. **4B** by arrow **436**. The green light initially encounters a portion of the spectral router **430** positioned above the first green split-pixel **404**, which passes the green light to the green spectral router **428** of the first green split-pixel **404**. Further, a portion of the spectral router **430** positioned above the red split-pixel **402** is configured to direct incident light in the green wavelength range to the green spectral router **428** of the first green split-pixel **404**. Consider, for purposes of discussion, green light entering the red split-pixel **402**. An example of green light is illustrated in FIG. **4B** by arrow **438**. The green light initially encounters a portion of the spectral router **430** positioned above the red split-pixel **402**, which directs the green light to the green spectral router **428** of the first green split-pixel **404**.

[0044] The red spectral router **426** is one example of a “first spectral router.” The red spectral router **426** is configured to direct incident light in a first subset (or sub-band) of the red wavelength range to the first red photodetector **410**. For example, the portion of the red spectral router **426** positioned above the first red photodetector **410** is configured to pass incident light in a first subset of the red wavelength range. Consider, for purposes of discussion, 660 nanometer light entering the red spectral router **426** above the first red photodetector **410**. An example of 660 nanometer light is illustrated in FIG. **4B** by arrow **440**. The 660 nanometer light initially encounters a portion of the red spectral router **426** positioned above the first red photodetector **410**, which directs the 660 nanometer light to the first red photodetector **410**. Further, a portion of the red spectral router **426** positioned above the first red photodetector **410** is configured to direct incident light outside of the first subset of the red wavelength range to one or more neighboring red photodetectors. Consider, for purposes of discussion, 700 nanometer light entering the red spectral router **426** above the first red photodetector **410**. An example of 700 nanometer light is illustrated in FIG. **4B** by arrow **442**. The 700 nanometer light initially encounters a portion of the red spectral router **426** positioned above the first red photodetector **410**, which directs the 700 nanometer red light to the second red photodetector **412**.

[0045] The red spectral router **426** is also configured to direct incident light in a second subset (or sub-band) of the red wavelength range to the second red photodetector **412**. For example, the portion of the red spectral router **426** positioned above the second red photodetector **412** is configured to pass incident light in a second subset of the red wavelength range. Consider, for purposes of discussion, 705 nanometer light entering the red spectral router **426** above the second red photodetector **412**. An example of 705 nanometer light is illustrated in FIG. **4B** by arrow **444**. The 705 nanometer light initially encounters a portion of the red spectral router **426** positioned above the second red photodetector **412**, which directs the 705 nanometer light to the second red photodetector **412**. Further, a portion of the red spectral router **426** positioned above the second red photodetector **412** is configured to direct incident light outside of the second subset of the red wavelength range to one or more neighboring red photodetectors. Consider, for purposes of discussion, 670 nanometer light entering the red spectral router **426** above the second red photodetector **412**. An example of 670 nanometer light is illustrated in FIG. **4B** by arrow **446**. The 670 nanometer light initially encounters a portion of the red spectral router **426** positioned above the second red photodetector **412**, which directs the 670 nanometer red light to the first red photodetector **410**.

[0046] In some implementations, the red spectral router **426** is configured to split red light into four subsets of the red wavelength range. For example, FIG. **5A** is a graph illustrating the red wavelength range. FIG. **5A** also illustrates an example of four subsets of the red wavelength range that the red spectral router **426** may split red light into. FIG. **5B** is a view of an example of the red split-pixel **402**. As shown in FIG. **5B**, the red spectral router **426** may direct incident light in a first subset of the red wavelength range to the first red photodetector **410**, direct incident light in a second subset of the red wavelength range to the second red photodetector **412**, direct incident light

in a third subset of the red wavelength range to the third red photodetector **414**, and direct incident light in a fourth subset of the red wavelength range to the fourth red photodetector **416**. In some implementations, the red spectral router **426** is configured to split red light into four subsets with substantially the same size. For example, the red spectral router **426** may direct incident light with wavelengths between 625 nanometers and 656 nanometers to the first red photodetector **410**, direct incident light with wavelengths between 657 nanometers and 688 nanometers to the second red photodetector **412**, direct incident light with wavelengths between 689 nanometers and 720 nanometers to the third red photodetector **414**, and direct incident light with wavelengths between 721 nanometers and 750 nanometers to the fourth red photodetector **416**. In other implementations, the red spectral router **426** is configured to split red light into four subsets with different sizes. For example, the red spectral router **426** may direct incident light with wavelengths between 625 nanometers and 639 nanometers to the first red photodetector **410**, direct incident light with wavelengths between 640 nanometers and 687 nanometers to the second red photodetector **412**, direct incident light with wavelengths between 688 nanometers and 735 nanometers to the third red photodetector **414**, and direct incident light with wavelengths between 736 nanometers and 750 nanometers to the fourth red photodetector **416**.

[0047] In some implementations, the red spectral router **426** is configured to split red light into two subsets of the red wavelength range. For example, FIG. 6A is a graph illustrating the red wavelength range. FIG. 6A also illustrates an example of two subsets of the red wavelength range that the red spectral router **426** may split red light into. FIGS. 6B, 6C, and 6D are views of examples of the red split-pixel **402** arranged diagonally, horizontally, and vertically. For example, as shown in FIG. 6B, the red spectral router **426** may direct incident light in a first subset of the red wavelength range to the first red photodetector **410** and the fourth red photodetector **416**, and direct incident light in a second subset of the red wavelength range to the second red photodetector **412** and the third red photodetector **414**. FIG. 6B is an example in which the first red photodetector **410** is positioned parallel to the fourth red photodetector **416** and the second red photodetector **412** is positioned parallel to the third red photodetector **414**.

[0048] In some implementations, the red spectral router **426** is configured to split red light into two subsets with substantially the same size. For example, the red spectral router **426** may direct incident light with wavelengths between 625 nanometers and 687 nanometers to the first red photodetector **410** and the fourth red photodetector **416**, and direct incident light with wavelengths between 688 nanometers and 750 nanometers to the second red photodetector **412** and the third red photodetector **414**. In other implementations, the red spectral router **426** is configured to split red light into two subsets with different sizes. For example, the red spectral router **426** may direct incident light with wavelengths between 625 nanometers and 700 nanometers to the first red photodetector **410** and the fourth red photodetector **416**, and direct incident light with wavelengths between 701 nanometers and 750 nanometers to the second red photodetector **412** and to third red photodetector **414**.

[0049] The green spectral router **428** is an example of a “third spectral router.” The green spectral router **428** is configured to direct different sub-wavelengths within the green wavelength range to different photodetectors in the first green split-pixel **404**. A similar discussion regarding the second green split-pixel **406** and the blue split-pixel **408**, each of which may be configured in a same or similar manner, is omitted so as not to unduly lengthen the specification.

[0050] Conventional hyperspectral image sensors require multiple exposures to generate multispectral images. However, the image sensor controller **214** is configured to determine the amount of incident light detected by each photodetector in each split-pixel of the pixel array **210** during each exposure. Thus, the image sensor **106** generates multispectral images from a single exposure and is faster than conventional sensors. Further, the image sensor controller **214** may combine the amounts of light detected by different photodetectors to determine color information. For example, the green photodetectors in the first green split-pixel **404** and the second green split-

pixel **406** that detect a subset in the upper portion of the green wavelength range may detect some yellow light. Thus, the image sensor controller **214** may determine an amount of yellow light by combining: (i) the amounts of light detected by photodetectors in the first green split-pixel **404** and the second green split-pixel **406** that are configured to detect a subset in the upper portion of the green wavelength range; and (i) the amount of light detected by the photodetector in the red split-pixel **402** that is configured to detect a subset in the lower portion of the red wavelength range. [0051] In some implementations, the image sensor controller **214** combines the amounts of light detected by only some of the photodetectors in a split-cell. For example, with reference to the configuration of the red split-pixel **402** discussed above in reference to FIGS. 5A and 5B, the first red photodetector **410** may detect some orange light because the subset of the red wavelength range detected by the first red photodetector **410** is close to the orange wavelength range. Thus, when characterizing red light detected by the red split-pixel **402**, the image sensor controller **214** may ignore the amount of light detected by the first red photodetector **410** and only use the amounts of light detected by the other three red photodetectors in the red split-pixel **402**.

[0052] In some implementations, the image sensor controller **214** is configured to determine a total amount of incident light in a color wavelength range detected by a split-pixel by combining the amounts of light detected by each of the four photodetectors in the split-pixel. For example, the image sensor controller **214** may determine a total amount of red light detected by the red split-pixel **402** by combining the amounts of light detected by each of the four photodetectors in the red split-pixel **402**. Thus, in addition to performing hyperspectral imaging, the image sensor **106** may also be used as a conventional RGGB sensor without having to change the structure of the image sensor pixels **212**.

[0053] FIG. 7 is a flow diagram of an example of a method **700** for constructing an image sensor pixel in accordance with some implementations. For simplicity of explanation, the method **700** is depicted in FIG. 7 and described as a series of operation. However, the operations can occur in various orders and/or concurrently, and/or with other operations not presented and described herein. At block **702**, a plurality of photodetectors is formed. The plurality of photodetectors includes at least a first photodetector and a second photodetector. For example, the red split-pixel **402** is formed with a plurality of photodetectors including at least the first red photodetector **410** and the second red photodetector **412**. At block **704**, a first spectral router is formed. The first spectral router is positioned above at least the first photodetector and the second photodetector. For example, the red spectral router **426** is formed above at least the first red photodetector **410** and the second red photodetector **412**. The first spectral router is configured to direct incident light in a first subset of a first color wavelength range to the first photodetector. For example, the red spectral router **426** is configured to direct incident light in a first subset of the red wavelength range to the first red photodetector **410**. The first spectral router is also configured to direct incident light in a second subset of the first color wavelength range to the second photodetector. For example, the red spectral router **426** is configured to direct incident light in a second subset of the red wavelength range to the second red photodetector **412**. At block **706**, a second spectral router is formed. The second spectral router is positioned above the first spectral router. For example, spectral router **430** is formed over the red spectral router **426**. The second spectral router is configured to direct the incident light in the first subset and the second subset of the first color wavelength range to the first spectral router. For example, spectral router **430** is configured to direct incident light in the first subset and the second subset of the red wavelength range to the red spectral router **426**. The second spectral router is also configured to direct incident light in a second color wavelength range to one or more neighboring image sensor pixels. For example, spectral router **430** is configured to direct incident light in the green wavelength range to the first green split-pixel **404** and the second green split-pixel **406**.

[0054] Many of the electrical connections in the drawings are shown as direct couplings having no intervening devices, but not expressly stated as such in the description above. Nevertheless, this

paragraph shall serve as antecedent basis in the claims for referencing any electrical connection as “directly coupled” for electrical connections shown in the drawing with no intervening device(s). [0055] The above discussion is meant to be illustrative of the principles and various implementations of the present invention. Numerous variations and modifications will become apparent to those skilled in the art once the above disclosure is fully appreciated. It is intended that the following claims be interpreted to embrace all such variations and modifications.

Claims

1. An image sensor pixel, comprising: a plurality of photodetectors including at least a first photodetector and a second photodetector; a first spectral router positioned above at least the first photodetector and the second photodetector, wherein the first spectral router is configured to: direct incident light in a first subset of a first color wavelength range to the first photodetector, and direct incident light in a second subset of the first color wavelength range to the second photodetector; and a second spectral router positioned above the first spectral router and configured to: direct the incident light in the first subset and the second subset of the first color wavelength range to the first spectral router, and direct incident light in a second color wavelength range to one or more neighboring image sensor pixels.
2. The image sensor pixel of claim 1, wherein the plurality of photodetectors further includes a third photodetector and a fourth photodetector, wherein the first spectral router is further positioned above the third photodetector and the fourth photodetector, and wherein the first photodetector, the second photodetector, the third photodetector, and the fourth photodetector are arranged in a two-by-two grid.
3. The image sensor pixel of claim 2, wherein the first spectral router is further configured to: direct incident light in a third subset of the first color wavelength range to the third photodetector, and direct incident light in a fourth subset of the first color wavelength range to the fourth photodetector.
4. The image sensor pixel of claim 2, wherein the first spectral router is further configured to: direct the incident light in the second subset of the first color wavelength range to the third photodetector, and direct the incident light in the first subset of the first color wavelength range to the fourth photodetector.
5. The image sensor pixel of claim 4, wherein the first photodetector is positioned parallel to the fourth photodetector, and wherein the second photodetector is positioned parallel to the third photodetector.
6. The image sensor pixel of claim 1, wherein the first color wavelength range is for a red color, and wherein the second color wavelength range is for a green color.
7. An image sensor comprising: a pixel array comprising an image sensor pixel including: a first photodetector, a second photodetector, a first spectral router configured to: direct incident light in a first subset of a first color wavelength range to the first photodetector, and direct incident light in a second subset of the first color wavelength range to the second photodetector, and a second spectral router configured to: direct incident light in the first color wavelength range to the first spectral router, and direct incident light in a second color wavelength range to one or more neighboring image sensor pixels in the pixel array; and a controller configured to: determine a first amount of light detected by the first photodetector, and determine a second amount of light detected by the second photodetector.
8. The image sensor of claim 7, wherein the image sensor pixel is a first image sensor pixel, wherein the pixel array further comprises a second image sensor pixel including: a third photodetector, a fourth photodetector, and a third spectral router configured to: direct incident light of a first subset of the second color wavelength range to the third photodetector, and direct incident light of a second subset of the second color wavelength range to the fourth photodetector, wherein

the second spectral router is further configured to route incident light of the second color wavelength range to the third spectral router.

9. The image sensor of claim 8, wherein the controller is further configured to: determine a third amount of light detected by the third photodetector, determine a fourth amount of light detected by the fourth photodetector, and determine an amount of a third color of light by combining the first amount of light and the third amount of light.

10. The image sensor of claim 7, wherein the image sensor pixel further includes a third photodetector and a fourth photodetector, and wherein the controller is further configured to: determine a third amount of light detected by the third photodetector, and determine a fourth amount of light detected by the fourth photodetector.

11. The image sensor of claim 10, wherein the controller is further configured to determine a total amount of light in the first color wavelength range by combining on the first amount of light, the second amount of light, the third amount of light, and the fourth amount of light.

12. The image sensor of claim 10, wherein the first spectral router is further configured to: direct incident light of a third subset of the first color wavelength range to the third photodetector, and direct incident light of a fourth subset of the first color wavelength range to the fourth photodetector.

13. The image sensor of claim 10, wherein the first spectral router is further configured to: direct the incident light of the second subset of the first color wavelength range to the third photodetector, and direct the incident light of the first subset of the first color wavelength range to the fourth photodetector.

14. The image sensor of claim 13, wherein the first photodetector is positioned diagonally adjacent to the fourth photodetector, and wherein the second photodetector is positioned diagonally adjacent to the third photodetector.

15. The image sensor of claim 7, wherein the first color wavelength range is between 625 nanometers and 750 nanometers, and wherein the second color wavelength range is between 495 nanometers and 570 nanometers.

16. The image sensor of claim 7, wherein the imaging sensor is included in at least one selected from the group consisting of an automobile, a vehicle, a camera, a cellular telephone, a tablet computing, a webcam, a video camera, a video surveillance system, and a video gaming system.

17. A method for constructing an image sensor pixel, the method comprising: forming a plurality of photodetectors including at least a first photodetector and a second photodetector; forming a first spectral router positioned above at least the first photodetector and the second photodetector, wherein the first spectral router is configured to: direct incident light in a first subset of a first color wavelength range to the first photodetector, and direct incident light in a second subset of the first color wavelength range to the second photodetector; and forming a second spectral router positioned above the first spectral router, wherein the second spectral router is configured to: direct the incident light in the first subset and the second subset of the first color wavelength range to the first spectral router, and direct incident light in a second color wavelength range to one or more neighboring image sensor pixels.

18. The method of claim 17, wherein the plurality of photodetectors further includes a third photodetector and a fourth photodetector, wherein the first spectral router is further positioned above the third photodetector and the fourth photodetector, and wherein the first photodetector, the second photodetector, the third photodetector, and the fourth photodetector are arranged in a two-by-two grid.

19. The method of claim 18, wherein the first spectral router is further configured to: direct incident light of a third subset of the first color wavelength range to the third photodetector, and direct incident light of a fourth subset of the first color wavelength range to the fourth photodetector.

20. The method of claim 18, wherein the first spectral router is further configured to: direct the incident light of the second subset of the first color wavelength range to the third photodetector,

and direct the incident light of the first subset of the first color wavelength range to the fourth photodetector.
